

Merrivoss Bioanalysis Inc

NM08 Paired Liver Systems

Microsomal CLint and Hepatocyte depletion CLint — Final report R1

Study MV-2006-090

Accession M06/BIO/02077

Prepared for Nimbravelle Medicines Inc

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1 Study disposition

The NM08 test article set comprised 24 compounds. The study opened on 06/05/2006, an interim interpretation was issued on 06/26/2006, and this report closes the paired liver-system work on 07/18/2006. Human and rat matrices are reported separately; neither species is substituted for the other in the progression assessment.

Table 1. Study identification and scope

Metric	Value
Project	MV-2006-090
Client	Nimbravelle Medicines Inc
Compounds tested	24
Series / chemotype	Nimbravelle aminotriazine series 08 (Aminotriazine)
Assays	Liver microsomal stability, Hepatocyte stability
Turnaround (calendar days)	43
Price	USD 19,200

Nimbravelle advanced the analogues whose slower hepatocyte turnover agreed with their microsomal ranking. This is a relative in vitro selection, not a prediction of measured circulating exposure.

2 Principal microsomal results

Table 2 gives Microsomal CLint at 0.5 mg/mL protein and 1 μ M test article. Clearance is expressed in μ L/min/mg; the associated depletion half-life is expressed in minutes. Human and rat columns must be read within their respective matrix.

Table 2. Microsomal CLint and depletion half-life

Compound	Human CLint (μ L/min/mg)	Human $t_{1/2}$ (min)	Rat CLint (μ L/min/mg)	Rat $t_{1/2}$ (min)
NM08-08406	47	29	68	21
NM08-08410	28	49	65	21
NM08-08416	5.9	>60	9.2	>60
NM08-08418	24	59	23	>60
NM08-08428	<3	>60	4.0	>60
NM08-08440	5.0	>60	10	>60
NM08-08452	5.7	>60	7.6	>60
NM08-08454	15	>60	16	>60
NM08-08460	7.7	>60	7.2	>60
NM08-08472	4.9	>60	8.5	>60
NM08-08476	3.4	>60	11	>60
NM08-08478	7.6	>60	15	>60
NM08-08496	6.5	>60	13	>60
NM08-08500	9.6	>60	12	>60
NM08-08502	7.2	>60	14	>60
NM08-08508	5.1	>60	6.2	>60
NM08-08518	7.7	>60	17	>60
NM08-08526	61	23	167	8.3
NM08-08548	13	>60	14	>60
NM08-08556	9.7	>60	22	>60
NM08-08566	6.2	>60	7.9	>60
NM08-08574	24	57	50	28
NM08-08578	<3	>60	7.3	>60
NM08-08590	5.0	>60	9.1	>60

Controls: verapamil (high), diltiazem (medium) within historical range

Half-life and intrinsic clearance describe the same depletion experiment and are not independent confirmation of activity. The matrix basis differs from the cell-normalized results that follow.

3 Principal hepatocyte results

Table 3 presents Hepatocyte depletion CLint in human and rat cryopreserved hepatocytes, at 1 μ M and 0.5×10^6 cells/mL. Results are expressed in μ L/min/ 10^6 cells. Comparison with Table 2 is by within-system ranking rather than subtraction of the printed clearance values.

Table 3. Hepatocyte depletion CLint by species

Compound	Human CLint (μ L/min/ 10^6 cells)	Rat CLint (μ L/min/ 10^6 cells)
NM08-08406	6.2	11
NM08-08410	11	20
NM08-08416	14	20
NM08-08418	11	28
NM08-08428	14	16
NM08-08440	15	24
NM08-08452	11	22
NM08-08454	5.3	12
NM08-08460	13	17
NM08-08472	24	51
NM08-08476	28	30
NM08-08478	4.4	11
NM08-08496	11	23
NM08-08500	16	30
NM08-08502	11	22
NM08-08508	20	36
NM08-08518	10.0	18
NM08-08526	25	85
NM08-08548	14	29
NM08-08556	21	31
NM08-08566	23	51
NM08-08574	47	154
NM08-08578	13	22
NM08-08590	15	38

A slower whole-cell depletion profile supports retention within this test article set when its direction agrees with the microsomal assessment. Whole-cell turnover also depends on access to the intracellular compartment.

4 Purpose and boundaries

The exposure gate addressed whether the aminotriazine analogues retained a consistent metabolic stability advantage across an isolated enzyme-rich matrix and an intact cellular matrix. The design does not isolate a particular metabolic enzyme or establish the pathway responsible for parent loss.

No plasma binding, transport, solubility or in vivo exposure result is supplied by this study. Those properties cannot be inferred from a favorable clearance ranking. The report therefore confines its recommendation to the paired liver systems.

5 Test article identity and accession

The controlling accession is M06/BIO/02077. Compound identifiers printed in Tables 2 and 3 are the link between the analytical record and the NM08 test article set. The accompanying assay worksheet holds the accession-lot mapping; identifiers must be carried intact into any later exposure-gate record.

The report uses the submitted parent identity as its analyte designation. It does not replace identity or purity characterization. An apparently favorable depletion profile cannot resolve a mismatch between the submitted material and its intended structure.

6 Matrix and species definition

Table 4. Matrix basis used throughout this report

System	Species	Normalization	Interpretation boundary
Liver microsomes	Human and rat separately	Protein mass	Enzyme-rich subcellular depletion
Cryopreserved hepatocytes	Human and rat separately	Cell number	Whole-cell parent depletion

Microsomal clearance is protein-normalized and hepatocyte clearance is cell-normalized. Conversion between these bases would require additional physiological scaling assumptions. None is applied here. A human result is not a quantitative rat surrogate, even when the direction of ranking agrees.

7 Microsomal procedure

Microsomal incubations used 0.5 mg/mL protein and 1 μ M test article. Timed aliquots were quenched for parent analysis. The depletion assessment considers the interpretable interval of the parent signal and the corresponding matrix controls, rather than treating every terminal sample as informative.

Loss unrelated to metabolic turnover must be considered when interpreting the series. Protein association, handling loss and matrix-dependent signal response can contribute to apparent depletion. Microsomal CLint is reported as an operational result under the stated incubation conditions.

8 Hepatocyte procedure

Hepatocyte incubations used 1 μM test article and 0.5×10^6 cells/mL. Human and rat preparations were evaluated independently. Parent depletion was normalized to the cell basis used in Table 3; that basis is retained in the captions and in the conclusion.

The intact-cell experiment incorporates entry into cells as well as intracellular turnover. Slow apparent clearance can reflect restricted access as well as metabolic stability. Agreement with microsomal ordering supports the selection but does not remove this limitation.

9 Bioanalytical measurement

Javier Quintana's mass-spectrometry review covers the parent-analyte measurement used for the depletion interpretation. The analyte card and assay worksheet remain the controlling records for sample assignment, acquisition details and integration decisions. This report presents the derived clearance tables rather than a reproduction of individual chromatograms.

Matrix-associated response changes must be distinguished from time-dependent loss of parent. A signal that cannot be interpreted within the analytical response range is not evidence of complete metabolic disappearance.

10 Analytical limits

The supplied result record does not provide numerical quantitation limits, calibration range or individual terminal-sample concentrations. Those values are therefore not asserted in this report. The tables support comparison of derived clearance; they do not support a claim that every terminal sample was quantifiable.

Where depletion approaches the measurement floor, late samples may cease to support the fitted interval. Where little parent loss occurs, the observation window limits resolution of slower turnover. Both constraints matter when interpreting close rankings, and neither justifies adding precision beyond the reported table.

11 Matrix acceptance

Helen Marchwood's matrix acceptance review is linked to the assay worksheet dated 07/18/2006. Acceptance is considered by system and species. A satisfactory record in human microsomes cannot release rat hepatocyte samples or resolve an unrelated matrix response concern.

- Retain accession-lot identity through sample assignment.
- Read the depletion interval alongside the applicable matrix record.
- Keep species and normalization basis visible in the released result.
- Escalate ambiguous parent loss before using it as a progression argument.

12 Acceptance exceptions and qualifications

No compound-specific acceptance exception is identified in the supplied final result tables. This statement is limited to the report dataset: it does not constitute an independent assertion that every underlying analytical sample passed an unreported numerical criterion.

The analytical-limit qualifications in Section 10 apply to the entire interpretation. The report does not invent sample exclusions, repeat runs or a numerical acceptance threshold. Any subsequent correction to the controlling assay worksheet should be assessed for its effect on the relevant species ranking before this report is reused.

13 Calculation assumptions

The depletion approach interprets parent loss over time under the stated matrix loading. Intrinsic clearance is an incubation-derived descriptor. Its comparison assumes consistent treatment of analyte identity, interpretable signal interval and matrix normalization across the test article set.

The half-life columns in the microsomal table must remain paired with their corresponding species clearance columns. Rounding can obscure small differences. It would be inappropriate to infer an additional significant digit, or to re-rank close entries using a rounded half-life alone.

14 Human-system interpretation

The human columns provide the primary paired-system comparison for this exposure gate. Read the human microsomal ordering first, then ask whether the human hepatocyte profile supports the same direction. A favorable result in only one system calls for qualification rather than an automatic stability claim.

The conclusion records the client's selection of analogues with agreement between slower hepatocyte turnover and microsomal ranking. It does not supply a new numerical pass threshold or identify a potency trade-off, because neither is contained in this study record.

15 Rat-system interpretation

Rat results provide a separate species context for the NM08 set. They should be retained alongside human findings when planning later exposure work. Concordance can support a coherent follow-up rationale; discordance should remain visible rather than being averaged away.

These rat in vitro tables do not establish rat exposure, oral availability or dose selection. Any later in vivo study would need its own route, formulation, sampling and bioanalytical record. No such result is included in this final report.

16 Cross-system comparison

A direct arithmetic comparison of $\mu\text{L}/\text{min}/\text{mg}$ with $\mu\text{L}/\text{min}/10^6$ cells has no defined meaning without a scaling model. The paired-system decision instead asks whether the relative direction of parent turnover agrees. The separate result tables preserve the evidence needed to revisit that judgment.

Agreement strengthens the case for retaining an analogue within this test article set; it does not prove a common rate-limiting process. Disagreement can be scientifically useful, indicating that cellular access or additional metabolic capacity may matter to interpretation.

17 Progression decision

Nimbravelle advanced the analogues whose slower hepatocyte turnover agreed with their microsomal ranking. Megan Larkin records that decision as the study outcome for Nimbravelle Medicines Inc. The decision rests on the matrix-specific tables and the qualifications stated above.

Later exposure-gate discussions should cite MV-2006-090 and preserve the original compound identifiers. The present recommendation is not a guarantee of systemic exposure or a replacement for the further characterization required by the client.

18 Study record and report control

This R1 final report is dated 07/18/2006. The related MV-2006-090_AssayWorksheet_071806_v03.xls contains the accession-lot mapping, matrix acceptance checks and paired assay results. The final report is the narrative interpretation; the worksheet is the accompanying structured assay record.

The interim communication of 06/26/2006 is superseded for final interpretation by this report. Corrections should retain the study and accession identifiers and explicitly state whether the released ranking or progression decision changes.

19 Assumptions retained with the report

- Parent depletion is interpreted under the incubation conditions shown with each result table.
- Species are evaluated separately.
- Microsomal and hepatocyte units are not interchangeable.
- No physiological scaling or exposure prediction is applied.
- Unreported quantitation limits and sample-level acceptance outcomes are not supplied by inference.

These assumptions travel with any extract from the result tables. Removing them would broaden the apparent claim beyond the work performed.

Name	Role	Signature	Date
Megan Larkin	DMPK Study Director	<u>M Larkin</u>	07/18/2006
Javier Quintana	Mass-Spectrometry Manager	<u>J Quintana</u>	07/18/2006
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